FOOD DELIVERY SYSTEM

DATA STRUCTURES AND ALGORITHMS MINI PROJECT REPORT

Submitted By

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ABSTRACT

The Food Delivery System is a console-based application developed using C++ and Data Structures. The system simulates the basic operations of an online food delivery platform such as managing food items, placing orders, processing deliveries, searching food items, sorting menus, and maintaining restaurant information.

Various data structures such as Arrays, Linked Lists, Stack, Queue, and Binary Search Trees have been used to efficiently manage and organize data. Searching and sorting algorithms have also been implemented to improve system performance.

The project helps students understand the practical applications of Data Structures and Algorithms in real-world systems.

INTRODUCTION

Food delivery applications have become an essential part of modern life. These platforms manage large amounts of information including restaurants, food items, customer orders, and deliveries.

The objective of this project is to create a simplified food delivery system using C++ while applying core concepts of Data Structures and Algorithms.

The project provides hands-on experience with different data structures and demonstrates how they can be used to solve practical problems efficiently.

OBJECTIVES

1. To implement a Food Delivery System using C++.
2. To understand practical applications of Data Structures.
3. To implement searching and sorting algorithms.
4. To manage customer orders efficiently.
5. To simulate delivery scheduling and restaurant management.
6. To analyze the time complexity of different operations.

SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux / MacOS
- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: GCC / G++

DATA STRUCTURES AND ALGORITHMS USED

1. Array (Vector)

Purpose:
Store food items including ID, name, and price.

Operations:

- Add Food Item
- Display Food Menu

Complexity:

- Insertion: O(1)
- Traversal: O(n)

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2. Linked List

Purpose:
Store customer orders.

Operations:

- Add Order
- Display Orders

Complexity:

- Insertion: O(n)
- Traversal: O(n)

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3. Stack

Purpose:
Undo the last order operation.

Principle:
LIFO (Last In First Out)

Operations:

- Push

- Pop
- Top

Complexity:
O(1)

---

4. Queue

Purpose:
Process deliveries in the order they were received.

Principle:
FIFO (First In First Out)

Operations:

- Enqueue
- Dequeue

Complexity:
O(1)

---

5. Binary Search Tree (BST)

Purpose:
Store and display restaurant information.

Operations:

- Insert Restaurant
- Inorder Traversal

Complexity:

- Average Insert: O(log n)
- Average Search: O(log n)

---

6. Linear Search

Purpose:
Search food items by name.

Complexity:
O(n)

---

7. Binary Search

Purpose:
Search food items by ID.

Complexity:
O(log n)

---

8. Insertion Sort

Purpose:
Sort food items according to price.

Complexity:
$O(n^2)$

SYSTEM FLOW

1. User adds food items.
2. Food items are stored in a vector.
3. Customer places an order.
4. Orders are stored in a linked list.
5. Order IDs are pushed into a stack for undo functionality.
6. Orders are inserted into a queue for delivery scheduling.
7. Users can search food items using Linear Search or Binary Search.
8. Food items can be sorted using Insertion Sort.
9. Restaurants are stored in a BST.
10. Delivery boy is assigned to the order.

MODULE DIVISION

Member 1

- Arrays (Vector)
- Food Item Management
- Linked List
- Order Management

Member 2

- Stack
- Undo Last Order
- Queue
- Delivery Scheduling

Member 3

- Display Food Menu
- Linear Search
- Binary Search
- Insertion Sort
- Complexity Analysis

Member 4

- Binary Search Tree
- Restaurant Management
- Delivery Boy Assignment

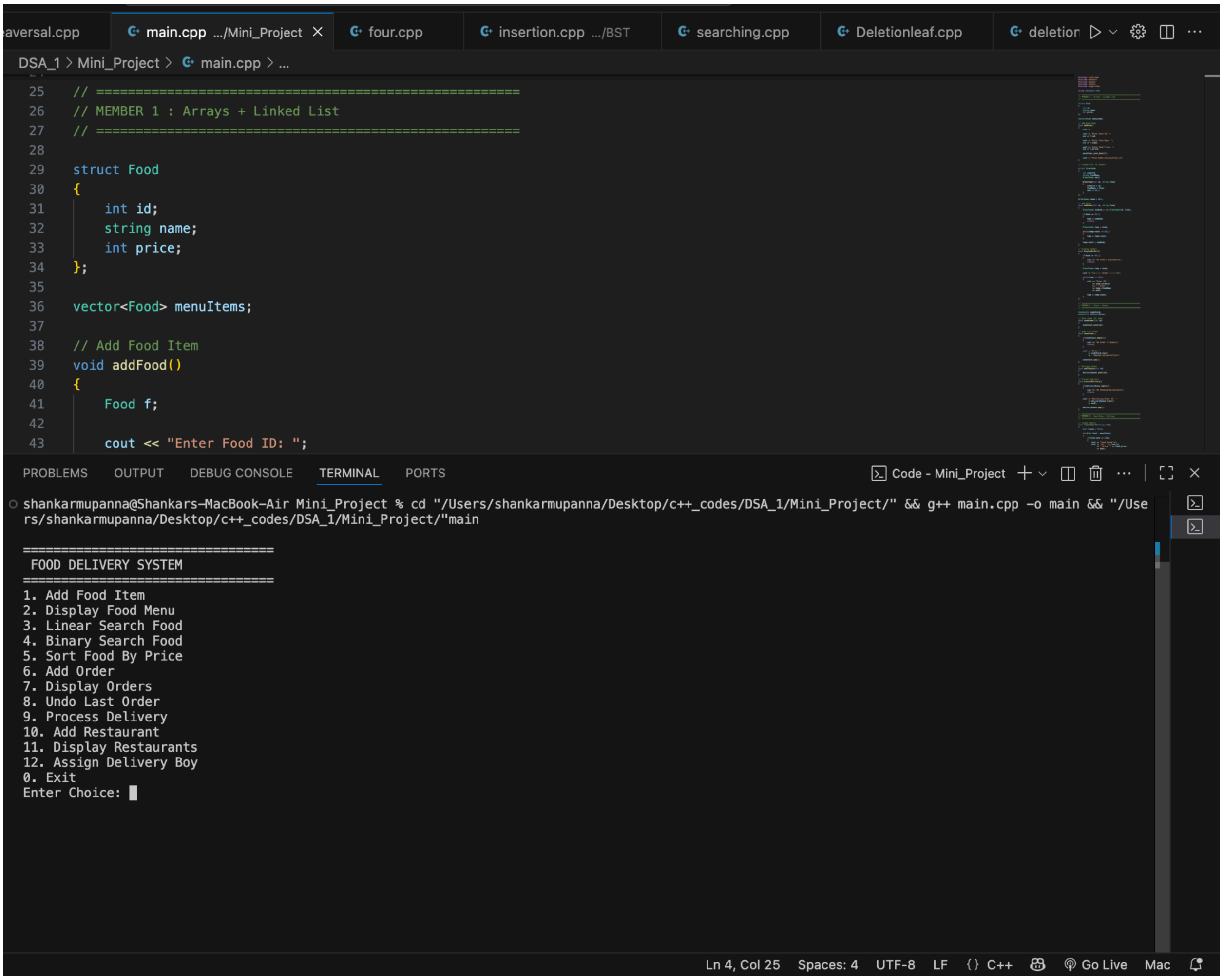
TIME COMPLEXITY ANALYSIS

Operation	Complexity
Add Food Item	O(1)
Display Food Menu	O(n)

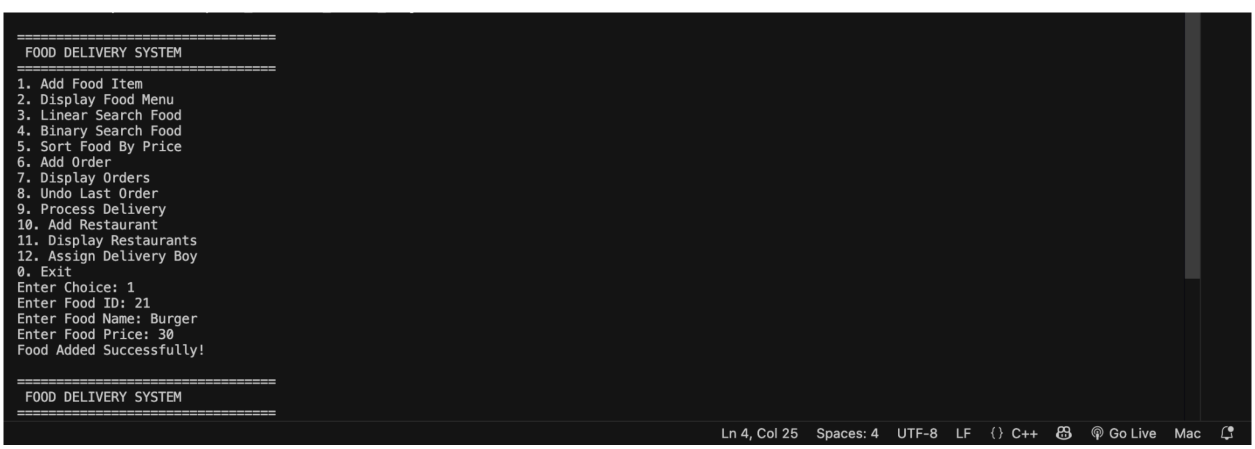
Linear Search	O(n)
Binary Search	O(log n)
Insertion Sort	O(n^2)
Add Order	O(n)
Stack Push	O(1)
Stack Pop	O(1)
Queue Push	O(1)
Queue Pop	O(1)
BST Insert	O(log n)
BST Traversal	O(n)

SAMPLE OUTPUT

Screenshot 1: Main Menu



Screenshot 2: Adding Food Items



Screenshot 3: Displaying Food Menu

```
0. Exit
Enter Choice: 2

------ FOOD MENU ------
21 | Burger | Rs.30

================================
 FOOD DELIVERY SYSTEM
================================
1. Add Food Item
2. Display Food Menu
3. Linear Search Food
4. Binary Search Food
5. Sort Food By Price
6. Add Order
7. Display Orders
8. Undo Last Order
9. Process Delivery
10. Add Restaurant
11. Display Restaurants
12. Assign Delivery Boy
0. Exit
Enter Choice: 
```

```
Ln 4, Col 25   Spaces: 4   UTF-8   LF   {} C++   Go Live   Mac
```

Screenshot 4: Searching Food Item

```
Food Added Successfully!

=================================
 FOOD DELIVERY SYSTEM
=================================
1. Add Food Item
2. Display Food Menu
3. Linear Search Food
4. Binary Search Food
5. Sort Food By Price
6. Add Order
7. Display Orders
8. Undo Last Order
9. Process Delivery
10. Add Restaurant
11. Display Restaurants
12. Assign Delivery Boy
0. Exit
Enter Choice: 4
Sorted Using Insertion Sort
Enter Food ID To Search: 23
Food Found
Burger Price = 30

=================================
```

Screenshot 5: Sorting Food Menu

```
=================================
 FOOD DELIVERY SYSTEM
=================================
1. Add Food Item
2. Display Food Menu
3. Linear Search Food
4. Binary Search Food
5. Sort Food By Price
6. Add Order
7. Display Orders
8. Undo Last Order
9. Process Delivery
10. Add Restaurant
11. Display Restaurants
12. Assign Delivery Boy
0. Exit
Enter Choice: 5
Sorted Using Insertion Sort
```

Screenshot 6: Adding Orders

```
=================================
 FOOD DELIVERY SYSTEM
=================================
1. Add Food Item
2. Display Food Menu
3. Linear Search Food
4. Binary Search Food
5. Sort Food By Price
6. Add Order
7. Display Orders
8. Undo Last Order
9. Process Delivery
10. Add Restaurant
11. Display Restaurants
12. Assign Delivery Boy
0. Exit
Enter Choice: 6
Order ID: 44
Food Name: Chapati
Order Added
```

Screenshot 7: Delivery Processing

```
=================================
 FOOD DELIVERY SYSTEM
=================================
1. Add Food Item
2. Display Food Menu
3. Linear Search Food
4. Binary Search Food
5. Sort Food By Price
6. Add Order
7. Display Orders
8. Undo Last Order
9. Process Delivery
10. Add Restaurant
11. Display Restaurants
12. Assign Delivery Boy
0. Exit
Enter Choice: 12
Assigned Delivery Boy : Rahul
```

ADVANTAGES

1. Easy to use.
2. Demonstrates multiple DSA concepts.
3. Modular implementation.
4. Efficient searching and sorting operations.
5. Real-world application of data structures.

LIMITATIONS

1. Console-based interface.
2. No database connectivity.
3. No user authentication.
4. Data is not permanently stored.

FUTURE ENHANCEMENTS

1. Graphical User Interface (GUI).
2. Database integration.
3. Online payment support.
4. Real-time order tracking.
5. Customer login and registration.

CONCLUSION

The Food Delivery System successfully demonstrates the implementation of important Data Structures and Algorithms in a practical application.

The project helped us understand how Arrays, Linked Lists, Stack, Queue, Binary Search Trees, Searching, and Sorting algorithms can be applied to solve real-world problems efficiently. It also improved our understanding of time complexity analysis and software development practices.

REFERENCES

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4. Cplusplus.com
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